

Step 2 — what the work is, and what to automate

Generated by `src/make_step2_report.py` from `out/segments.jsonL` . Do not edit by hand — an earlier hand-maintained version silently went three pipeline revisions out of date.

All figures are **relative**. The brief states the recordings were made in a test environment with compressed waiting times, so absolute durations are meaningless and only comparisons between processes are valid.

Scale

664 executions, 173 minutes of observed work, 15 sessions, 4 operators, all on 2026-07-01.

How many people

source	count
username_hash	4
machine_id	4
operator names on the portal dashboard	3

Where each of those names appears. The names themselves are withheld; each row is keyed by the system the name belongs to.

name belongs to	machines it appears on	sessions it appears in	appearances in that system
fin	4 of 4	15 of 15	100%
hr	4 of 4	15 of 15	100%
ops	4 of 4	13 of 15	100%

The three names are **not** operators — each appears on every machine, and each belongs to one portal system. They are shared per-system logins. So: **4 people** working across three systems under shared accounts, which is a governance finding rather than a trivial one — automation would run with no per-user audit trail.

The route names describe nothing

The portal is one SPA deployed three times, so its route names repeat. The regulation document open during the work identifies it instead:

segment label	dominant document	segments	purity
ops__leave-applications	keiyaku_kaijo_tetsuzuki	52	65%
hr__onboarding	nyusha_checklist_shinsotsu_batch	51	80%
fin__payroll-items	getsujitsu_teigaku_torihikisaki_ichiran	50	94%
fin__onboarding	gyomu_itaku_kyuuyo_kitei	30	97%
fin__leave-applications	settai_keihi_kitei	27	85%
fin__resident-tax	shinkuitorihikisaki_touroku_tetsuzuki	20	90%
hr__payroll-items	gyomu_itaku_keihi_kitei	18	56%
hr__social-insurance	ikuji_kyuugyou_kitei	17	59%
ops__social-insurance	kanrisya_kengen_shinsei_tetsuzuki	7	100%
hr__leave-applications	keiyaku_kaijo_tetsuzuki	2	50%

Weighted document purity across labels: **80.4%**. The pipeline never reads document names, so this is independent evidence that the groupings track real business processes.

Ranking

Two measurable quantities in tension: **mechanical load** (clipboard events and application switches per run — what automation removes) against **judgment load** (share of runs with a regulation document open — what it does not). High volume with high judgment is a poor first target, because the residue is what costs the time.

Priority is the hand transfers automation would remove, discounted for judgment: $\text{executions} \times \text{transfers per run} \times (1 - \text{judgment share})$. The table is in that order.

process	n	min	share	median	mech	judgment	people
payroll_item_maintenance	120	27.2	15.8%	11 s	2.6	15%	4
inventory_payroll_items	63	15.5	9.0%	11 s	3.1	3%	4
new_supplier_registration	73	15.6	9.0%	11 s	2.1	27%	4
leave_application_review	58	13.2	7.7%	10 s	2.0	3%	3
recurring_supplier_payment	77	23.4	13.6%	16 s	3.9	65%	4
admin_privilege_request	42	8.5	4.9%	11 s	2.7	17%	4
childcare_leave_handling	35	7.7	4.5%	13 s	3.0	49%	3
new_grad_onboarding	58	19.2	11.1%	19 s	6.2	88%	4
contract_termination	59	19.1	11.1%	19 s	6.2	88%	4
contractor_payment_setup	37	11.0	6.4%	17 s	5.1	81%	3
entertainment_expense_approval	34	9.5	5.5%	16 s	4.9	79%	2
fin_social_insurance	8	2.7	1.6%	21 s	3.5	0%	1

Robustness

A priority formula is easy to make say what you want, so the ranking was scored under 5 different weightings. Processes with $n < 5$ were excluded, since per-run rates on one or two observations are noise.

process	top-3 appearances
payroll_item_maintenance	5 / 5
recurring_supplier_payment	2 / 5
new_grad_onboarding	2 / 5
new_supplier_registration	2 / 5
inventory_payroll_items	2 / 5
leave_application_review	1 / 5
contract_termination	1 / 5

Only **payroll_item_maintenance** survives every weighting. Anything appearing once is an artefact of a particular formula, not a finding.

The result that sets the scope

Aggregating by portal **screen** rather than by system. Judgment is the share of the screen's runs with a regulation document open:

screen	executions	minutes	share of all work	systems	judgment
payroll-items	260	66.1	38.3%	3	27%
leave-applications	151	41.8	24.2%	3	54%
onboarding	95	30.1	17.5%	2	85%
social-insurance	85	18.9	11.0%	3	28%
resident-tax	73	15.6	9.0%	1	27%

One screen pattern accounts for 38.3% of all observed work, occurs in 3 systems, and carries the lowest judgment load of the five screens, though only just: 26.9%, against 27.4% for resident-tax and 28.2% for social-insurance. The two top-ranked processes are this one screen in different deployments.

That reframes Step 3: the choice is not *which process to automate* but *one bespoke process, or the shared pattern behind several* — which is exactly the scope question the brief says is itself part of the ROI decision.